

SANJIT MATHUR

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Education & Summary

University of Wollongong in Dubai – Expected 2027

Computer and Autonomous Systems Engineering student with experience building ML systems and LLM-powered tools in fintech and aviation environments. Deployed containerized services on Kubernetes and built predictive models achieving 88% accuracy on operational datasets.

Skills

Languages: Python, TypeScript, C, SQL, MATLAB

AI/ML: LLM APIs (prompt engineering, RAG), OpenCV, scikit-learn, Pandas, NumPy, Matplotlib, Stacking Ensembles, Quantile Regression

Backend: Node.js, Express.js, REST APIs, Pydantic (Data Validation)

Infrastructure & Tools: Docker, Kubernetes, PostgreSQL, Streamlit, Plotly, Joblib

Frontend: React, Next.js

Experience

AI Engineering Intern | Baraka Financial Ltd., Dubai (Feb 2026 – Present)

- Deployed containerized services to Kubernetes; built AI-powered internal tooling that automated error classification and log analysis, eliminating manual triage across support workflows.
- Built Position Search, Trading Account Monitor, CRA, EOD History, Slack Automation and other modules integrating OMS, Instruments, and Wallet microservices, reducing manual portfolio lookup time for operations teams.
- Unified inconsistent data schemas across distributed services into a single portfolio state model, eliminating data discrepancies in cross-market account views.

Digital Intern | IndiGo InterGlobe Aviation Ltd., Gurgaon (August 2025 – Sept 2025) – [GitHub](#)

- Built a Logistic Regression model to predict on-time arrival performance on the DEL-BOM sector using 1,000 flight records and 6 engineered features (block-hour overrun, departure delay, aircraft type, weather index, ATC congestion), achieving 88% accuracy with balanced precision/recall across both classes.
- Engineered features from raw operational data including one-hot encoding of 3 aircraft types (A320, A320neo, A321neo), computed block-hour overrun from scheduled vs. actual hours, and conducted correlation analysis across 10 numerical variables to inform feature selection for modeling.

Software Engineering Intern | Lab of Future, Dubai (Jun 2025- Aug 2025)

- Built internal educational software used by 500+ students across 4 campuses.
 - Designed backend APIs for course content delivery and student progress tracking.
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Projects

ExamForge - [GitHub](#), [Website Link](#)

- Built a full-stack AI-powered exam generation platform using LLM APIs for dynamic multi-format question creation.
- Designed interactive workflows allowing students to generate, attempt, and review customized assessments.

Multi Domain Demand Forecaster - [GitHub](#)

- Built demand forecasting engine using XGBoost + LightGBM stacking ensemble with Ridge meta-learner for cross-domain trend prediction.
- **Uncertainty Quantification & Feature Engineering:** Implemented **Quantile Regression** to provide 10th/90th percentile confidence intervals and engineered complex temporal features, including cyclical sine/cosine encodings for seasonality and domain-specific rolling statistics.
- **Full-Stack ML Deployment:** Built and deployed a RESTful **FastAPI** service for real-time batch predictions and an interactive **Streamlit** dashboard for multi-domain data visualization and model performance tracking.

Orvyn ExoArm – Rehabilitation Exoskeleton (Current – May 2026)

- Developing signal classification pipeline to differentiate between intended finger movements and noise, targeting 90%+ accuracy for assisted rehab exercises.
- Designing a microcontroller-based rehabilitation exoskeleton that interprets sEMG signals to assist finger movement through real-time signal processing and controlled servo actuation.

F1 Simulation Dashboard - [GitHub](#)

- Built a full-stack race simulation engine modeling tire degradation, pit strategy, safety cars, weather, and overtaking across all 24 2026 GPs using Monte Carlo probability methods.
- Implemented dual-mode interface: season-wide predictions and fully configurable custom race simulations. -

